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Key active mercury methylating microorganisms and their synergistic effects on methylmercury production in paddy soils

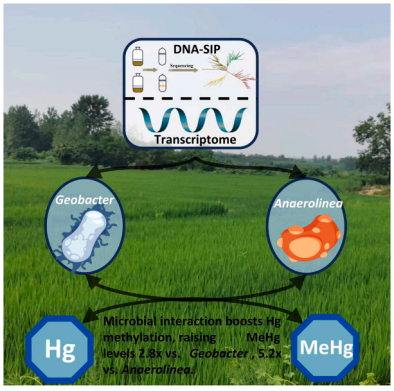
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HIGHLIGHTS

- *Geobacter* and *Anaerolinea* were identified as key Hg methylators in paddy soils.
- Synergy of *Geobacter* and *Anaerolinea* boosts MeHg production in paddy soils.
- Study provides insight for predicting soil MeHg load and rice contamination risks.

GRAPHICAL ABSTRACT



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ABSTRACT

Rice contamination with neurotoxic methylmercury (MeHg) from paddy soils is an escalating global concern. Identifying the microorganisms responsible for mercury (Hg) methylation in these soils is essential for controlling Hg contamination in the food chain and mitigating health impacts. Current research often focuses on total Hg-methylating microorganisms, overlooking the contributions of active ones, which can lead to either overestimating or neglecting the specific roles of microorganisms in Hg methylation within paddy soils. In this study, active Hg-methylating microorganisms in paddy soils were identified using a combination of DNA-SIP, Hg isotope labelling, and Hg methylation gene sequencing techniques. Our findings revealed that *Geobacter* and *Anaerolinea* are pivotal active Hg-methylating microorganisms across a contamination gradient in paddy soils. Transcriptomic analysis of soils from major rice-producing provinces in China confirmed the widespread and synergistic involvement of these microorganisms. Microbial incubation further validated their interaction significantly enhances Hg methylation, with $Me^{198}Hg$ concentrations increasing 2.8-fold compared to *Geobacter* alone and 5.2-fold compared to *Anaerolinea* alone. These findings enhance our understanding of microbial Hg

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methylation in paddy soils, providing critical insights for accurately predicting soil MeHg load, rice grain MeHg contamination, and human MeHg exposure risks.

1. Introduction

The organic form of mercury (Hg)-Methylmercury (MeHg) can be bioaccumulated in food chains in aquatic and wetland ecosystems. This bioaccumulation negatively affects human, animal, and ecosystem health [1,2]. Fish, seafood, and marine mammals are recognized as the primary pathways for human MeHg exposure [3,4]. Recent studies have also identified rice consumption as a significant route of MeHg exposure for residents especially in Hg-contaminated areas [5,6]. Over the past few decades, global attention has increasingly focused on MeHg exposure from rice consumption [7], as reports of MeHg-contaminated rice have emerged in major rice-exporting countries like China [5,6], Indonesia [8], India [9], Pakistan [10], and the Philippines [11]. Rice paddy soil is a known hotspot for MeHg production [12,13], which result to the accumulation of MeHg in rice grains throughout the rice growing season [14-16]. Therefore, a comprehensive understanding of the factors influencing MeHg production in paddy soils is critically needed.

Although organic matter [17], redox potential [18], and the biogeochemical cycling of elements, such as iron [19], sulfur [20], and nitrogen [21], all influence Hg methylation in paddy soils, their impact is largely attributable to the effects on the activity and abundance of Hg-methylating microorganisms. Consequently, Hg-methylating microorganisms play a crucial role in Hg methylation in paddy soils [20,22]. Sulfate-reducing bacteria (SRB) have been proposed as significant contributors to MeHg production in paddy soils. This is supported by evidence that inhibiting SRB with $MoSO_4^{2-}$ often decreases Hg methylation [23], and the abundance of the SRB biomarker (*dsrB* gene) is significantly correlated with MeHg concentration [24,25]. The discovery of genes essential for Hg methylation (*hgcAB* gene) has enabled more efficient, standardized, and precise monitoring of Hg methylators in various environments [26]. Consequently, microbial Hg-methylating communities, including SRB, iron-reducing bacteria (IRB), methanogens, and syntrophs, have recently been characterized in paddy soils [27,28]. However, the key Hg-methylating microorganisms in paddy soils remain inadequately documented, with discrepancies even within the same study area [29-31]. For example, in the paddy soils of Wanshan, Tongren, China, Vishnivetskaya et al. [30] identified sulfate-reducing δ-proteobacteria and methanogenic archaea as potential Hg-methylating microorganisms. Liu et al. [29] demonstrated the involvement of IRB and methanogens in Hg methylation. Furthermore, Zhang et al. [31] reported that methanogenic archaea promote microbial Hg methylation. These divergent findings are likely due to the inclusion of many inactive Hg-methylating microorganisms in DNA-based sequencing analyses. Given the abundant presence of extracellular DNA in soil [32], DNA-based 16S rRNA gene sequencing, Hg methylation gene (i.e., *hgcA* gene) sequencing, and metagenomic sequencing identify not only active microorganisms but also inactive ones. In this context, "inactive" refers both to dead microorganisms and to live microorganisms that harbor the *hgcAB* genes but do not express them under specific environmental conditions [33]. The presence of these inactive microorganisms may lead to an overestimation or neglect of the role of certain microorganisms in Hg methylation, thereby preventing us from accurately predicting MeHg loads in paddy soils, MeHg accumulation in rice grains, and the associated human exposure risks. Therefore, it is imperative to conduct in-depth research on active Hg-methylating microorganisms in paddy soils.

Recent advancements have introduced several methods for identifying active microorganisms, each with its strengths and limitations. RNA-based methods, such as metatranscriptomics [34], focus on gene expression and functional potential but are limited by RNA instability [35]. Metaproteomics [36] measures protein profiles, offering direct activity insights, but is hindered by the complexity of protein extraction and reliance on extensive database [37]. Metabolomics provides a snapshot of metabolic activity but often lacks the taxonomic resolution needed to identify specific organisms [38]. In contrast, DNA-based Stable Isotope Probing (DNA-SIP) has gained prominence for its unique ability to link microbial activity directly to function by incorporating isotopically labeled substrates (e.g., ^{13}C or ^{15}N) into the DNA of active microorganisms [39-42]. This method has been successfully applied in various ecosystems, including soil [43], aquatic environments [44], and sediments [42], to accurately identify active microorganisms and connect them to specific metabolic processes, providing a functional perspective on microbial ecology. Despite the cost and meticulous experimental requirements of isotopically labeled substrates, DNA-SIP excels in its precision, effectively distinguishing active microbial populations from inactive or dead cells. This precision makes DNA-SIP an invaluable tool for gaining insights into microbial roles in biogeochemical cycles [45]. Therefore, DNA-SIP represents a valuable opportunity to identify active Hg-methylating microorganisms in paddy soils and stands out as the most ideal method for this purpose.

Thus, we employed a novel combination of DNA-SIP, Hg isotope labeling, and *hgcA* gene sequencing to pinpoint key Hg-methylating microorganisms in paddy soils across a contamination gradient. Transcriptomics was utilized to ascertain the relative importance and the synergistic involvement of these active Hg-methylating microorganisms in regulating Hg methylation in paddy soils from major rice-growing provinces across China, with further validation provided by the microbial incubation with Hg isotope. The insights gained from this study, particularly the identification of key active Hg-methylating microorganisms in paddy soils, offer a refined framework for assessing the net production of MeHg in paddy soils, the enrichment of MeHg in rice grains, and the associated human exposure risks.

2. Materials and methods

2.1. Site description and sample collection

In this study, we conducted two distinct sampling campaigns to comprehensively investigate Hg-methylating microorganisms in paddy soils. The first sampling campaign focused on specific areas within Guizhou Province, while the second expanded to major rice-producing provinces across China.

2.1.1. First sampling campaign

Paddy soils were gathered from three distinct areas within Guizhou Province, China. The first was an abandoned Hg mining area (AM), the second was an artisanal Hg smelting area (AS) within the Wanshan Hg mining district, and the third was a control area in Guiyang City (BG). These areas were chosen to represent different levels of Hg contamination, net MeHg production, and soil microbial community diversity [46]. In each area, three rice paddies were chosen at random, and one soil sample was collected from the root zone (10–20 cm depth) of each paddy field 60 days after rice transplantation in 2021, resulting in nine samples. Although these samples were not strictly rhizosphere soils, they were collected from the 10–20 cm depth around the root zone, an area characterized by active microbial processes and nutrient cycling due to interactions between roots and the surrounding soil [47]. This layer was chosen because it represents the root-adjacent environment where microbial activity is enhanced, and methylmercury uptake by rice primarily occurs through root absorption. Each sample was divided into two subsamples: one for SIP microcosm experiments and the other for

physicochemical properties, *hgcA* gene, and Hg analyses. Detailed information on the study area and sample collection for fist sampling campaign is available in SI Text S1.

2.1.2. Second sampling campaign

Paddy soils (root zone, 10–20 cm depth) were collected from major rice-producing provinces in China, including Jilin, Guangxi, Hunan, Guizhou, Zhejiang, Liaoning, and Shaanxi. In each province, soil samples were collected from four locations within a 500 m^2 area of the same rice field and mixed into one composite sample, totaling nine samples. These samples were collected 50–60 days after rice transplantation in 2022. Each sample was split into two subsamples: one preserved in liquid nitrogen for transcriptomic analysis of active mercury methylating microorganisms, and the other stored at ~4 °C for Hg analysis. Further information on the soil samples for second sampling campaign is provided in SI Table S1.

2.2. Experimental design

To systematically decipher the key active Hg-methylating microorganisms in paddy soils and their synergistic effects on MeHg production, three experiments were performed in this study.

2.2.1. Experiment 1

Identification of active Hg-methylating microorganisms was achieved through a SIP microcosm experiment. Microcosms were established using paddy soils from the first sampling campaign, utilizing inorganic Hg isotope as Hg donors [48] and acetate as methyl donors [49]. 30 mL of paddy slurry (prepared by adjusting the moisture content of paddy soils to 75 %, ref Liu et al. [20]) was purged with nitrogen to create an anoxic environment. The ^{202}Hg treatment was added with enriched $^{202}Hg(NO_3)_2$ stable isotope (with a spiked concentration of ^{202}Hg at 10 % of the background THg in soil, ref Hintelmann et al. [50]). A 10 mM concentration of either ^{13}C-labeled (99 %, Sigma-Aldrich, Shanghai, China) or ^{12}C-labeled (99 %, Sigma-Aldrich, Shanghai, China) acetate was used as a methyl donor in the ^{13}C-labeled (^{202}Hg-^{13}C[acetate]) and ^{12}C-labeled (^{202}Hg-^{12}C[acetate]) treatments, respectively. All the treatments were performed in triplicate and incubated in the dark at 25 °C for 3 days. Post-incubation, isotope-enriched MeHg ($Me^{202}Hg$) was measured using gas chromatography-inductively coupled plasma-mass spectrometry (GC-ICP-MS, Agilent 7700 ×, Agilent Technologies, Inc., USA) system using the ethylation-purge-trap method [50]. Genomic DNA of samples were extracted for active Hg-methylating microorganisms' identification. Notably, the possibility of the transformation of Hg^{2+} into MeHg via abiotic effects was ruled out in this study, as no discernable MeHg was generated from available inorganic Hg with acetate (more details are described in the SI Text S2).

2.2.2. Experiment 2

The importance and role of the pre-identified active Hg-methylation microorganisms were elucidated through transcriptomic analysis of field-collected samples. A total of nine soil samples were obtained from major rice-producing provinces across China during the second sampling campaign. Metatranscriptomic sequencing was employed to examine the functional roles and activity levels of the identified Hg-methylating microorganisms in real paddy soil environments, providing a broader understanding of their significance across different regions.

2.2.3. Experiment 3

Synergistic effects of pre-identified key Hg-methylating microorganisms (*Geobacter* and *Anaerolinea*) on MeHg production were achieved through anaerobic microbial incubation experiment. Anaerobic microbial incubation was conducted in serum bottles within an oxygen-free glovebox (PLASLABS, USA). Detailed cultivation methods for *Geobacter sulfurreducens* PCA and *Anaerolinea* sp. P10 are provided in SI Text S3. Incubation paddy slurries were prepared by mixing paddy soil from BG (first sampling campaign) and Milli-Q water. Each homogenized slurry (30 mL) was placed in a serum bottle with four treatments: Control, *Geobacter*-added, *Anaerolinea*-added, and a combined *Geobacter*- and *Anaerolinea*-added treatment. For the three microbial treatments, a standardized bacterial culture (10^6 CFU/g, ref Labana et al. [51]) was added to the respective treatments. All the bottles were spiked with $^{198}Hg(NO_3)_2$ tracer, at a dosage matching that used in the SIP experiment. The bottles were then sealed, gently shaken to mix the tracer and slurries, and covered with aluminum foil.

Incubation lasted 48 h at 25 °C in the glovebox. The incubation temperature was selected based on typical soil temperatures in paddy fields during the rice-growing season, which are conducive to the activity of both *Geobacter* and *Anaerolinea*. These microorganisms are mesophilic and are known to thrive in similar temperature ranges, making the selected temperature suitable for both. Post-incubation, the isotope-enriched MeHg species ($Me^{198}Hg$) was measured using GC-ICP-MS with the ethylation-purge-trap method [50].

2.3. DNA extraction, stable isotope probe gradient fractionation, hgcA gene sequencing and bioinformatic analysis

After completing the SIP microcosm experiment, genomic DNA was extracted from both ^{202}Hg-^{13}C[acetate] and ^{202}Hg-^{12}C[acetate] treatments using the FastDNA® Spin Kit (MP Biomedicals, Irvine, CA) according to the manufacturer's instructions. Approximately 800 ng of DNA was combined with CsCl solution to reach an initial buoyant density of 1.790 g/mL and then centrifuged at 130 000 g for 65 h at 20 °C. The stratified DNA-CsCl solution (around 400 μL) was divided into 12 fractions using an automatic sampler (model BSZ-100, Jiapeng, Shanghai, China) according to a published protocol [52]. The buoyant density of each fraction was measured using a refractometer (Atago, Tokyo, Japan), and the fractions were then purified following the published protocol [53]. DNA concentration in each fraction was measured using Qubit (Invitrogen, CA, USA). The *hgcA* gene was quantified using the primer sets ORNL-Delta-HgcA-F: GCCAACTACAAGMTGASCTWC and ORNL-Delta-HgcA-R: CCSGCNGCRCACCAGACRTT [29]. The selected DNA fractions were further processed for Hg methylation gene sequencing with the primers ORNL-HgcAB-uni-F (5′-AAYGTCTGGTGYGCNGCVGG-3′) and ORNL-HgcAB-uni-32R (5′-CAGGCNCCGCAYTCSATRCA-3′) adopted from Gionfriddo et al. [54].

Random forest analysis, using the "randomForest" package [55], was employed to predict the influence of active Hg-methylating microorganisms and physicochemical factors (e.g., dissolved organic matter, iron, and sulfur) on Hg methylation. Co-occurrence networks were constructed using the "igraph" [56] and "psych" [57] to identify key Hg-methylating microorganisms within the community based on strong (Spearman's $r > 0.8$) and significant ($p < 0.01$) correlations. Details for amplicon sequencing and sequence analysis are provided in SI Text S4.

2.4. RNA extraction, transcriptomic sequencing, and bioinformatic analysis

Approximately 10 g of each soil sample (second sampling campaign) underwent RNA extraction using the PowerSoil® Total RNA Isolation Kit (QIAGEN, Germany), following the manufacturer's protocol. The quality of the extracted RNA was assessed using a Nanodrop One Spectrophotometer (Thermo Fisher Scientific, USA) and verified through 1 % agarose gel electrophoresis, while RNA quantity was measured using a Qubit Fluorometer (Thermo Fisher Scientific, MA, USA).

High-quality RNA samples were used for transcriptome sequencing. cDNA libraries were constructed using the Illumina TruSeq RNA Sample Preparation Kit (Illumina, USA) and sequenced on the Illumina NovaSeq 6000 platform with 150 bp paired-end reads. Raw sequencing data underwent stringent quality control to eliminate adapter sequences and

low-quality reads, resulting in high-quality clean data. Clean reads were mapped against the NCBI Rfam database, and rRNA sequences were removed using sortmerna v.4.2.0 [58]. The remaining mRNA sequences were then subjected to de novo assembly using Trinity v.2.4.0 [59]. Post-assembly filtering ensured the selection of high-quality contigs for downstream analysis, including open reading frame (ORF) prediction, *hgcA* gene detection, function annotation (e.g., carbon cycling genes and Hg resistance genes) and taxonomic annotation. The details for the transcriptomic analysis are available in SI Text S5.

2.5. Analysis of hgcA gene, Hg and physicochemical properties

The abundance of the *hgcA* gene was quantified using the primer sets ORNL-Delta-HgcA-F: GCCAACTACAAGMTGASCTWC and ORNL-Delta-HgcA-R: CCSGCNGCRCACCAGACRTT on an AB 7500 instrument (Applied Biosystems, USA) [29]. Details of the amplification procedures are available in SI Text S6.

The pH and EC of the soil samples were measured using a pH electrode (SX751 pH/ORP/Cond/DO meter, SANXIN, China) with a soil: water ratio (w/v) of 1:2.5. Soil dissolved organic matter (DOM, reflected by water-soluble DOC) was extracted with Milli-Q water (soil: water = 1:10 w/v), filtered (0.45 μm polypropylene membrane filters) and determined using a total organic carbon analyzer (Vario TOC cube, Elementar, Germany). Soil sulfate (SO_4^{2-}) and nitrate (NO_3^-), were measured by ion chromatography (IC, ICS90, DIONEX, USA). The S^{2-} and Fe^{2+} in soil pore water were obtained by centrifugation (3500 g for 15 min) and measured using the methylene blue method [60] and the ferrozine method [61], respectively.

Total Hg in paddy slurries was digested by aqua-regia and determined by cold vapor atomic fluorescence spectrometry (CVAFS, Brooks Rand Model III, Brooks Rand Laboratories) [62]. MeHg in paddy slurries was digested by $CuSO_4$-methanol/solvent extraction and determined by GC-CVAFS [63]. Detailed information on Hg analyses is available in SI Text S7.

The isotope-enriched MeHg species ($Me^{202}Hg$ and $Me^{198}Hg$) in paddy slurries were measured by GC-ICP-MS with ethylation-purge-trap method [50]. Detailed information concerning the isotope-enriched MeHg measurement and the calculation of MeHg and methylation rate constant is described in SI Text S8.

2.6. Statistical analysis and quality control

Significance analyses of the raw data were conducting using SPSS 26.0 software (SPSS, Chicago, IL) with a threshold of $p < 0.05$ indicating statistical significance. Quality assurance and control (QA/QC) for mercury and isotopic mercury measurements included method blanks, triplicates, and certified reference materials. Certified reference materials (CRMs) GBW07405 (GSS-5, [THg] = 290 ± 40 ng/g) and ERM-CC580 ([MeHg] = 75.5 ± 3.7 ng/g) were employed as standards for the analyses of THg, MeHg, and $Me^{2o2}Hg$. Recovery rates for THg, MeHg, and $Me^{2o2}Hg$ were 92 ± 3 % (n = 10), 100.7 ± 10.9 % (n = 10), and 94 ± 7 % (n = 8), respectively. The method detection limits (3σ) for soil samples were 0.01 ng/g for THg, 0.002 ng/g for MeHg, and 0.05 ng/g for MeHg isotopes. Triplicate samples were included in every batch of

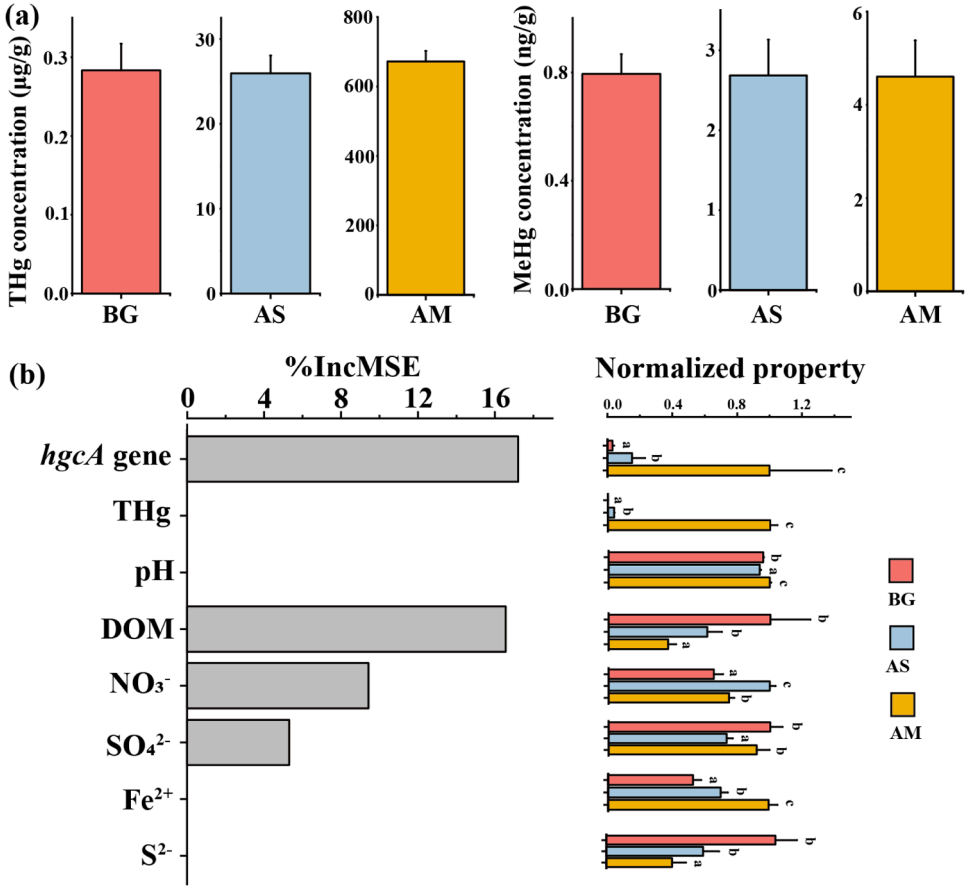


Fig. 1. The effect of mercury-methylating microorganisms on mercury methylation. (a) Total mercury and methylmercury concentration in different paddy soils. **(b)** Differences in physicochemical properties of different paddy soils and their effects on mercury methylation. Mercury methylation is represented by the MeHg concentration. Only predictors with significant effects are shown in the figures. BG, background area; AM, abandoned Hg mining area; AS, artisanal Hg smelting area. Significant differences among different treatments were tested with Tukey's honest significance test; different letters in each bar indicate significant differences among treatments ($p < 0.05$).

10 samples during the measurement process, with the relative standard deviation (RSD) for triplicates remaining below 7 % for all analytes.

3. Results

3.1. Variations in physicochemical properties, Hg content, and hgcA gene

As illustrated in Fig. 1**a**, the average THg concentration significantly increased from 0.28 ± 0.03 μg/g in the BG to 25.96 ± 2.09 and 672.05 ± 30.26 μg/g in the AS and AM, respectively. The MeHg concentrations mirrored the trend for THg, with mean concentrations of 0.79 ± 0.07 ng/g in BG, 2.68 ± 0.45 ng/g in AS, and 4.61 ± 0.78 ng/g in AM (Fig. 1**b**). Although certain physicochemical factors, such as THg, pH, NO_3, and Fe^{2+}, exhibited significant differences among the different paddy soils (BG, AS, and AM), random forest analysis revealed that the influence of *hgcA* gene abundance on Hg methylation was slightly greater than that of these physicochemical parameters (Fig. 1**c**).

Environmental variables were initially included to explore their potential correlations with Hg methylation activity. However, our analysis indicated that microbial factors, particularly *hgcA* gene abundance, had a more substantial impact on Hg methylation than environmental factors. As a result, these environmental variables were not further emphasized in the results and discussion. This finding reinforces the notion that microbial activity plays a dominant role in Hg methylation within paddy soils.

3.2. Identification of key Hg-methylating microorganisms by DNA-SIP

A significant increase in *hgcA* gene abundance was observed in the ^{202}Hg-^{13}C[acetate] and ^{202}Hg-^{12}C[acetate] treatments compared to the control (^{202}Hg) treatment (Fig. S1), indicating that acetate amendment could promote microbial MeHg production. This hypothesis was validated by the increased Me^{202}Hg concentration and Hg methylation rate (k_m-$^{202}Hg^{2+}$), which were higher in the acetate treatments than in the control (Fig. S2). In this incubation experiment, the observed difference between Me^{202}Hg concentration and k_m-$^{202}Hg^{2+}$ may be attributed to variations in microbial activity, as revealed by the fact that Raman spectroscopy analysis (Fig. S3), which showed that microbial activity is higher at low Hg concentrations compared to high Hg concentrations (see SI Text S9 for additional details). Additionally, no significant difference in k_m-$^{202}Hg^{2+}$ was observed between the ^{202}Hg-^{13}C[acetate] and ^{202}Hg-^{12}C[acetate] treatments, suggesting no preference for either ^{13}C[acetate] or ^{12}C[acetate] in MeHg formation.

Genomic DNA from the ^{202}Hg-^{13}C[acetate] and ^{202}Hg-^{12}C[acetate] treatments was separated into "heavy" and "light" fractions based on their buoyant density using isopycnic gradient centrifugation. Quantification of the *hgcA* gene revealed a distinct shift in distribution along the density gradient between the ^{202}Hg-^{13}C[acetate] treatment and ^{202}Hg-^{12}C[acetate] treatments (Fig. S4), suggesting that active Hg-methylating microorganisms were tagged during incubation. Therefore, to determine the community compositions of active Hg-methylating microorganisms, *hgcA* gene amplicons from the "heavy" fractions of ^{202}Hg-^{13}C[acetate] treatment and corresponding fractions of ^{202}Hg-^{12}C[acetate] treatment (shaded area, Fig. S4) were sequenced, as microorganisms highly enriched in fractions in ^{13}C-labeled treatment relative to ^{12}C control are functional microorganisms [52,64].

The microbial community responsible for Hg methylation and actively assimilating acetate was primarily composed of bacteria from

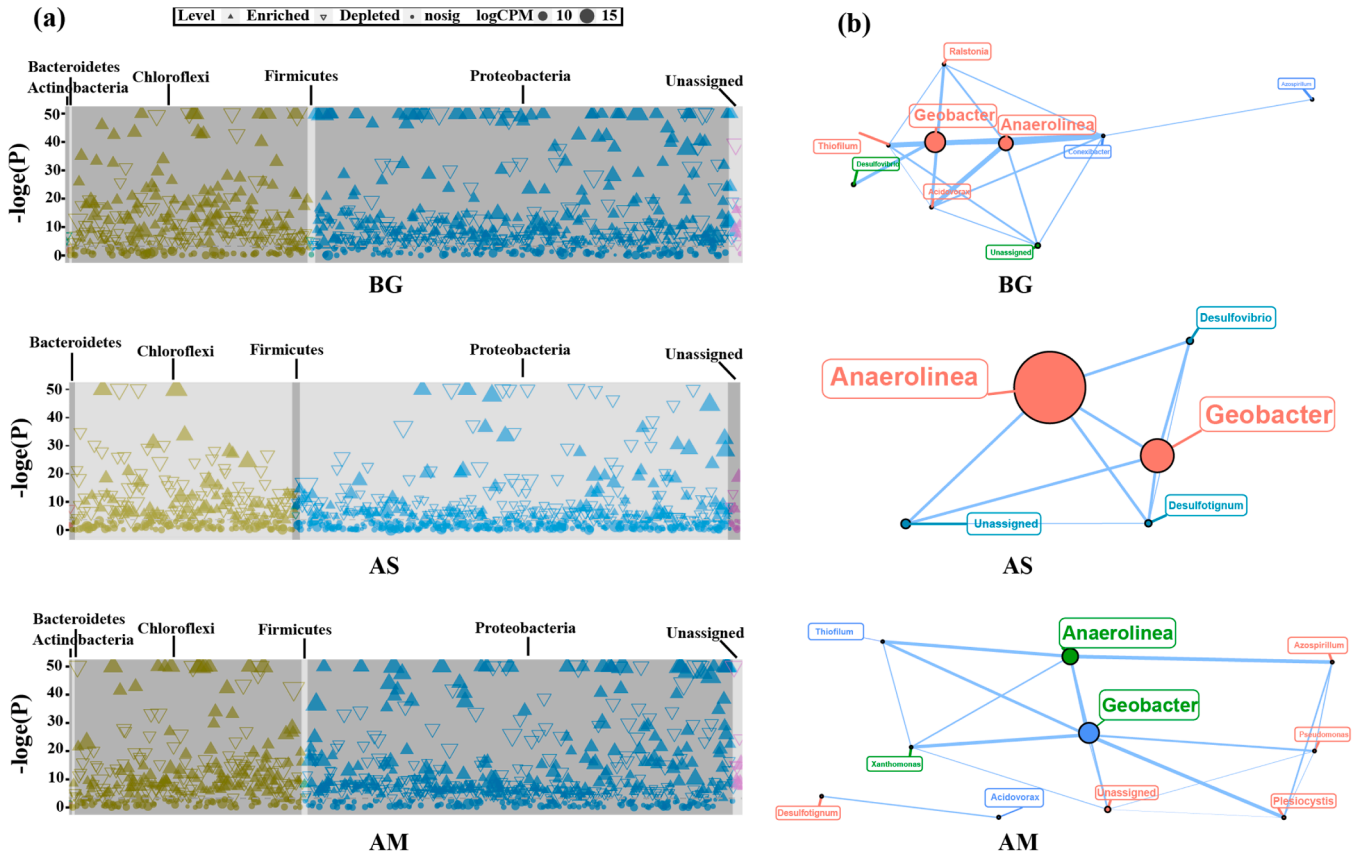


Fig. 2. Key Hg-methylating microorganisms in paddy soils. (a) Manhattan plot showing OTUs enriched in ^{13}C-labeled or ^{12}C-unlabeled treatments in paddy soils from the background area (BG), abandoned Hg mining area (AM), and artisanal Hg smelting area (AS). Each dot or triangle represents a single OTU. OTUs enriched in ^{13}C-labeled or ^{12}C-unlabeled samples are represented by filled or empty triangles, respectively. **(b)** Network analysis revealing co-occurrence patterns among Hg methylating microorganisms at genus level. Nodes are colored according to modularity classes. The size of each node is proportional to the number of connections.

the phyla *Proteobacteria*, *Chloroflexi*, *Firmicutes*, *Actinobacteria*, and *Bacteroidetes* (Fig. 2**a**). Specifically, the identified Hg-methylating microorganisms included *Geobacter* (65.63 %), *Anaerolinea* (17.51 %), *Desulfovibrio* (0.65 %), and *Terracoccus* (<0.01 %) in the BG; *Geobacter* (27.49 %), *Anaerolinea* (20.46 %), and *Desulfotignum* (0.04 %) in the AS; and *Geobacter* (61.20 %), *Anaerolinea* (25.64 %), *Desulfotignum* (0.04 %), and *Desulfovibrio* (0.50 %) in the AM (Fig. S5). *Geobacter* and *Anaerolinea* were recognized as the primary Hg methylators within the community due to their high number of connections with other microorganisms in the community (Fig. 2**b**) and their highest betweenness centrality (Fig. S6), indicating their pivotal roles in maintaining Hg-methylating community structure and function.

3.3. Transcriptional profiling of Hg-methylating microorganisms and their functional potential

To further validate the SIP results, we conducted transcriptomic analysis on soil samples collected from major rice-producing regions

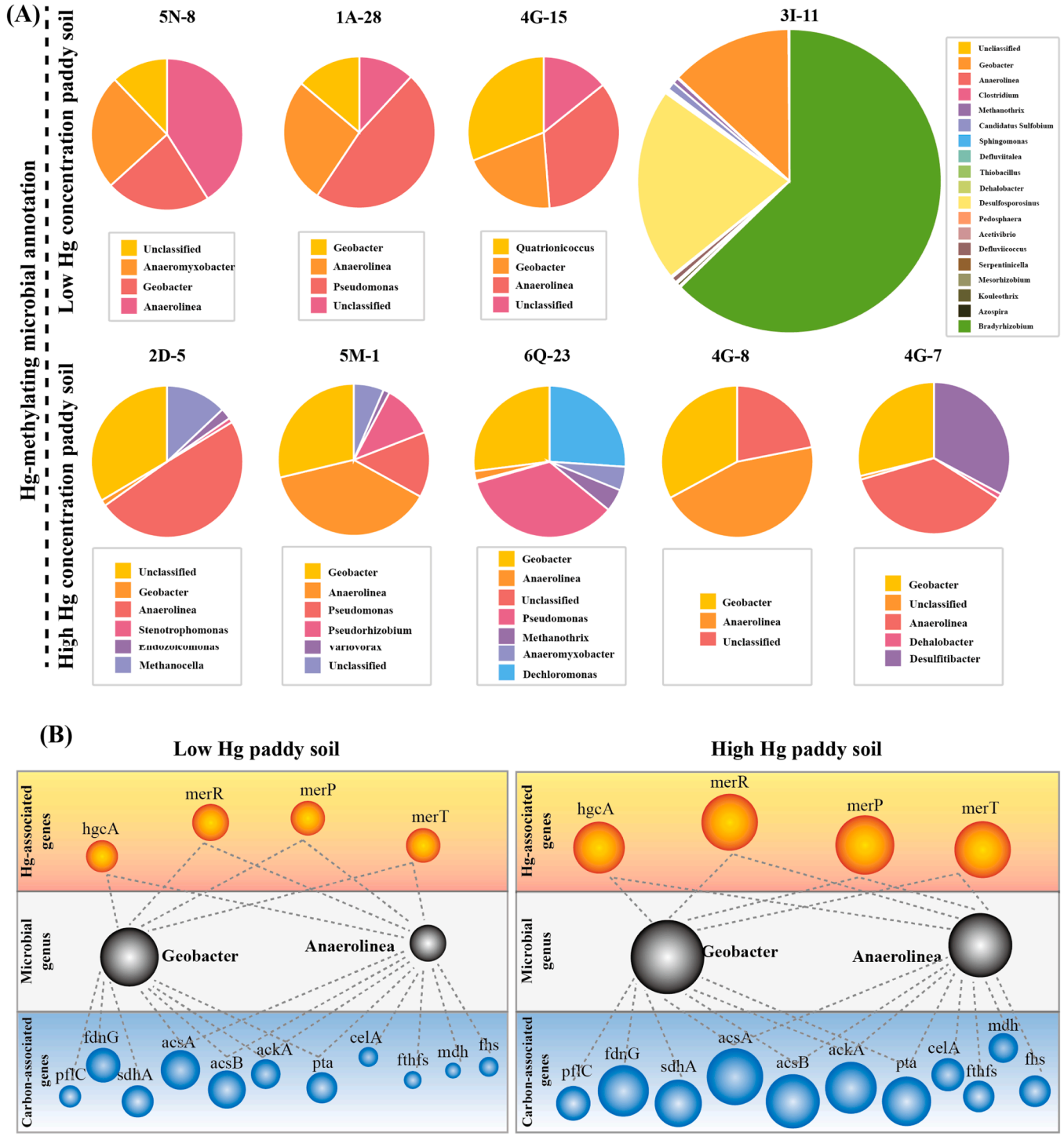


Fig. 3. Transcriptomic of Hg-methylating microorganisms in paddy soils. (a) The Composition and relative abundance of Hg-methylating microbial community in paddy soils. (b) Distribution of Hg-associated and carbon-associated genes in the active *Geobacter* and *Anaerolinea* within paddy soils. The area of the circles represents the abundance of bacteria and genes detected in the database.

across China. At the genus level, *Geobacter* and *Anaeroolinea* emerged as high-abundance Hg-methylating microorganisms, with their average abundances reaching 34 % and 23 %, respectively, in paddy soils (Fig. 3**a**). Notably, their relative abundance increased with higher Hg concentrations (Fig. 3**b**). Intriguingly, a synergistic relationship between *Geobacter* and *Anaerolinea* was observed, evidenced by the simultaneous increase in their abundances (Fig. S7). This finding suggests a potential mutualistic symbiosis between *Geobacter* and *Anaerolinea* in paddy soils.

Further analysis through KEGG database comparisons substantiated this hypothesis. *Anaerolinea* harbored genes such as *celA*, *ackA*, *pta*, *acsA*, *acsB*, *fthfs*, *mdh*, and *fhs* (Fig. 3**b**), indicating roles in acetyl-CoA synthesis, formate metabolism, and acetate metabolism [65-67]. *Geobacter*, on the other hand, possessed genes including *pflC, acsA, acsB, ackA, pta, fdnG,* and *sdhA* (Fig. 3**b**), associated with electron transfer, formate metabolism, and acetyl-CoA synthesis [65–68].

Furthermore, comparison with the BacMet database [69] revealed that both *Geobacter* and *Anaerolinea* carry the *merR*, *merP*, and *merT* genes (Fig. 3**b**). This indicates that these microorganisms poses Hg resistance gene [70], enabling them to survive and function effectively in Hg-contaminated paddy soils.

3.4. Impact of synergistic interactions among Geobacter and Anaerolinea on Hg methylation

To validate the transcriptomic results, incubation experiments were conducted with different microbial amendment treatments (*Geobacter*, *Anaerolinea*, and a combination of both) and a control treatment without microbial addition. Analysis of $Me^{198}Hg$ production from inorganic Hg isotope ($^{198}Hg^{2+}$) revealed that microbial amendment treatments produced higher amounts of $Me^{198}Hg$ compared to the control treatment (Fig. 4). The $Me^{198}Hg$ concentrations in the *Geobacter* and *Anaerolinea* amendment treatments were 2.77 ng/g and 1.50 ng/g, respectively, which are considerably lower than the 7.80 ng/g observed when both microorganisms were added together. This result confirms that the synergistic interaction between *Geobacter* and *Anaerolinea* significantly enhances Hg methylation in paddy soils, boosting the methylation process beyond what is achievable by each microorganism alone.

4. Discussion

In this study, the *hgcA* gene abundance was found to play a predominant role in the methylation process within Hg gradient paddy soils. This finding emphasizes the critical importance of microbial activity in the formation of MeHg in such environments [19]. Building upon this observation, our current study identified active *Geobacter* and *Anaerolinea* as key Hg-methylating microorganisms through DNA-SIP incubation with [^{13}C]-labeled acetate and sequencing of Hg methylation gene. These microorganisms exhibit the highest number of connections and betweenness centrality within the Hg-methylating microbial community network, which are critical indicators for identifying core microorganisms in the community [71].

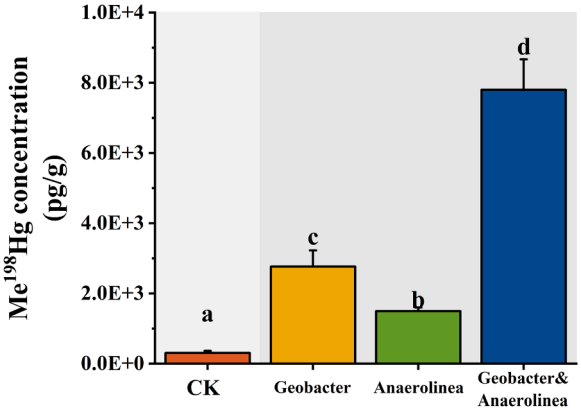


Fig. 4. Mercury methylation during the incubation across different treatments. CK, Control treatment; *Geobacter*, *Geobacter*-added treatment; *Anaerolinea*, *Anaerolinea*-added treatment; *Geobacter&Anaerolinea*, combined *Geobacter*- and *Anaerolinea*-added treatment. Significant differences among different treatments were tested with Tukey's honest significance test; different letters in each bar indicate significant differences among treatments ($p < 0.05$).

We acknowledge that not all Hg-methylating microorganisms in paddy soil rely on acetic acid metabolism pathway and therefore cannot be identified using labeled acetate. This limitation may lead to the neglect of the importance of other Hg-methylating microorganisms. Nevertheless, we believe that [^{13}C]-labeled acetate successfully identifies Hg-methylating microorganisms occupying significant ecological niches in paddy soil, as acetate is the main carbon source for most anaerobic microorganisms such as iron reducing bacteria, sulfur-reducing bacteria, and methanogens [72,73].

Transcriptome analysis further substantiated that *Geobacter* and *Anaerolinea* are widespread and highly abundant in paddy soils collected from major rice-growing provinces across China. This reinforces their significant role in Hg methylation and demonstrates the broad applicability of the DNA-SIP results across diverse geographical regions of China's major rice production areas. However, DNA-SIP based *hgcA* gene sequencing revealed that the abundance of key active Hg-methylating microorganisms did not increase with higher Hg concentrations, whereas transcriptomic analysis indicated an increase in their abundance with rising Hg levels. This discrepancy might be due to the varying sequencing techniques and data processing methods. The sensitivity and specificity of amplicon sequencing and metatranscriptomic analysis differ, potential leading to discrepancies in the detected microbial abundance. Regardless of the sequencing method employed, *Geobacter* and *Anaerolinea* were consistently identified as key active Hg-methylating microorganisms in paddy soils.

It should be noted that elevated Hg concentrations may serve as a selective pressure, favoring microorganisms with Hg-methylating capacity. However, this relationship is complex, as microbial abundance and activity are also shaped by factors such as redox conditions, electron donor availability, and competition among microbial taxa [74,75]. Raman spectroscopy results further revealed that the ratio values for indigenous microorganisms in paddy soils were lower at high Hg concentrations compared to low Hg concentrations. This indicates that microbial activity is higher under low mercury exposure conditions than under high mercury exposure conditions [76,77]. Thus, while elevated Hg concentrations may enrich Hg-methylating microorganisms, severe heavy metal contamination can suppress their metabolic activity. This explains why MeHg content can be high in heavily contaminated soils, despite a low Hg methylation rate.

Building on this understanding, we cannot simply assume that microorganisms with high abundance are important Hg-methylating microorganisms, as they do not necessarily demonstrate metabolic activity. Indeed, the presence of functional genes alone does not confirm their expression or activity under natural conditions. Therefore, our study employed a multi-faceted approach that integrates DNA-SIP, network analysis, transcriptomics, and microbial activity monitoring to comprehensively understand the key players in Hg methylation. Both *Geobacter* and *Anaerolinea* were confirmed to express the *hgcA* gene, although the *hgcB* gene was not detected, likely due to sequencing or assembly limitations [78]. Their active role in Hg cycling is further supported by the presence of the *mer* operon, which facilitate Hg^{2+} excretion from cells [70] and subsequent reduction of Hg^{2+} to Hg^0 through electron transport [79,80]. The *mer* operon-mediated microbial reduction of Hg^{2+} to Hg^0 has been proven to be an efficient detoxification mechanism [81].

Although the number of IRB with Hg methylation capacity is considerably lower than that of SRB [82], the Hg-methylating abilities of the IRB model strain *G. sulfurreducens* PCA are comparable to those of

Desulfovibrio desulfuricans ND132 (*D. desulfuricans* ND132) [83], suggesting the significance of IRB in MeHg formation [84]. Despite a recent study by Zhong et al. identifying *Geobacteraceae* as significant Hg-methylating microorganisms in paddy soil [85], their conclusions were speculative and lacked direct evidence to confirm the importance of *Geobacterace*, or to identify the specific microbial species or groups involved. Given that iron is the fourth most abundant element on the Earth's surface [86], its prevalence in rice fields suggests a potentially high abundance of iron-reducing bacteria. Consequently, the high abundance of iron-reducing bacteria may overshadow the contributions of other microorganisms in the Hg methylation process, particularly those less abundant yet impactful [87].

Our study supports this hypothesis and specifically identifies active *Anaerolinea* as key Hg-methylating microorganism in paddy soils, despite their lower abundance compared to *Geobacter*. It is well established that *Anaerolinea* are the dominant microorganisms that drive the sulfur and carbon cycles in natural environments [88,89]. These results all indicated that active *Anaerolinea* are also key Hg-methylating microorganisms in paddy soils, potentially promoting mercury methylation through Hg-sulfur coupling [20,90,91] or provide a carbon source for other Hg methylators [91]. These findings highlight the importance of considering a broader range of microorganisms in studies of Hg methylation.

Given that the abundance of microbial functional genes, rather than the overall microbial community structure, provides a better informative framework for understanding soil nutrient cycles [92], the abundance of key Hg-methylating microorganisms such as *Geobacter* and *Anaerolinea* can be used as proxies for the abundance of Hg methylation gene with the following prerequisites: (i) functional profiling of microbial communities is performed using Hg methylation gene sequencing, and (ii) *Geobacter* and *Anaerolinea* are keystone species in microbial communities. In summary, our study underscores the critical role of both *Geobacter* and *Anaerolinea* in the Hg methylation process in paddy soils.

Transcriptome analysis suggested a potential synergistic role in carbon cycling due to the presence of *acsA* and *acsB* genes in both *Anaerolinea* and *Geobacter*. While the presence of identical genes implies overlapping metabolic functions, it does not guarantee identical functional expression under natural conditions. Gene expression can vary based on intracellular regulation, nutrient availability, and environmental stressors [93,94]. Additionally, these microorganisms may compete for shared substrates, such as acetate, which could influence their relative abundance and activity. This competition may drive niche differentiation, encouraging specialization in distinct metabolic roles to reduce overlap [74].

Despite this potential competition, cooperative interactions are also possible. *Geobacter*'s iron-reducing capabilities create favorable redox conditions that enhance the metabolic activity of *Anaerolinea* in degrading complex organic compounds, such as cellulose and acetate [95]. This mutualistic relationship allows both microorganisms to coexist efficiently, facilitating nutrient cycling and contributing to mercury methylation in paddy soils.

The enhanced MeHg production resulting from the synergistic interaction of *Geobacter* and *Anaerolinea* indicates that microbial consortia play a crucial role in the biogeochemical cycling of mercury in paddy soils. This synergistic relationship not only increases MeHg concentration but also suggests that such microbial interactions are essential for the overall health and functionality of the ecosystem. Future research should aim to elucidate the precise molecular mechanisms underpinning this microbial synergy. Understanding these mechanisms could reveal new strategies for manipulating microbial interactions to control MeHg levels in contaminated environments, thereby mitigating mercury pollution and its harmful effects.

5. Conclusion

This study identified *Geobacter* and *Anaerolinea* as key Hg-methylating microorganisms in paddy soils, highlighting their synergistic role in enhancing MeHg production. Incorporating their abundances into Hg biogeochemical models could enhance the accuracy of risk assessments. The identification of *Anaerolinea* as a significant Hg-methylator expands our understanding of microbial Hg methylation beyond rice fields. These findings provide essential insights for managing MeHg production, particularly in the face of global climate change, and underscore the importance of further research on microbial interactions in Hg-sensitive ecosystems.

Environmental implication

This study identifies *Geobacter* and *Anaerolinea* as key active Hg-methylating microorganisms in paddy soils, revealing their synergistic role in significantly enhancing methylmercury (MeHg) production. By combining DNA-SIP, Hg isotope labeling, and gene sequencing, this research provides a novel understanding of microbial Hg methylation in contaminated rice-growing regions, offering critical insights for accurately predicting MeHg contamination in rice and associated human health risks. This work advances current knowledge by focusing on active Hg methylators rather than total microbial communities, which is crucial for developing targeted strategies to mitigate MeHg exposure through rice.

CRediT authorship contribution statement

Qiang Pu: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Kun Zhang:** Writing – review & editing, Formal analysis, Data curation. **Jiang Liu:** Writing – review & editing, Formal analysis, Conceptualization. **Qianshuo Zhang:** Writing – review & editing, Formal analysis, Data curation. **Mahmoud A. Abdelhafiz:** Writing – review & editing, Formal analysis, Data curation. **Bo Meng:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Xinbin Feng:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2024.136481.

Data Availability

Data will be made available on request.

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